

Online path planning for disaster response in the presence of unknown road disruptions: a comparison of research and practice

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Abstract

Purpose – This paper aims to illuminate gaps between theory and practice in one crucial area of disaster logistics: transportation management between staging areas and final destinations, on uncertain and disrupted road networks. Relevant current practice in federal, state, nonprofit and private disaster response organizations in the USA is described and compared to pertinent assumptions in the literature on disaster response online path planning under uncertainty.

Design/methodology/approach – The study uses a qualitative approach. The disaster response routing literature informed the development of an interview guide used for in-depth interviews with disaster response logistics practitioners. Collected data were coded to identify how path planning activities were accomplished. Interview findings were then linked back to the literature to identify where theory coincides and/or contrasts with practice.

Findings – The study identifies a number of important mismatches between current practice and the assumptions in the modeling literature. Current practice likely would not support online path planning models in which algorithms: run within drivers' vehicles, depend on knowledge of the probability of roads being open or closed, assume immediate information updates on road status or optimize speed at the expense of reliability. For research advancements to be leveraged, practitioners must adopt formal systems for sharing information, and research must adapt to the constraints and goals of response practitioners.

Originality/value – Despite the availability of sophisticated models to support disaster response path planning, few are adopted in practice. This research provides a better understanding of disaster transportation management and its challenges. The findings could help bring research into better alignment with the needs of practitioners, thereby improving the speed and effectiveness of the response and reducing the suffering of those affected by disasters.

Keywords Disaster relief transportation planning, Uncertain road conditions, Online models for path planning

Paper type Research paper

1. Introduction

Disasters are becoming more prevalent, due to the rapid environmental and socioeconomic changes happening on multiple fronts (Mitra and Shaw, 2023; The World Bank, 2023). The number of annual natural disasters has increased 10-fold from 1960 to 2019 (Institute for Economics and Peace, 2020), and could rise from approximately 400 in 2015 to 560 by 2030 [United Nations Office for Disaster Risk Reduction (UNDRR), 2022]. Economic consequences are also rising: over 380 billion-dollar climate and weather disasters have occurred in the USA since 1980, exceeding \$2.7tn in total cost (NCEI, 2024).

Disasters often cause shortages in essential utilities and resources like water, food, medical supplies and shelter. They also cause significant damage to buildings and critical infrastructures, such as homes, hospitals, power grids, bridges, road networks and water systems (FEMA, 2020). Logistics plays a crucial role in supporting disaster-impacted

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people, by providing critical supplies to affected areas and minimizing detrimental effects on infrastructure and people (Holguín-Veras *et al.*, 2012). Estimates indicate that 75–80% of the total investment in disaster response activities is attributed to logistics (Trunick, 2005; Stumpf *et al.*, 2023) and 40% of these investments represent waste associated with duplication of efforts and lack of detailed planning and analysis (Day *et al.*, 2012; Van Wassenhove 2006).

This paper studies one crucial area of disaster logistics: transportation between staging areas and final destinations on uncertain and damaged road networks. In the USA, a Federal Staging Area (FSA) serves as the logistical support hub for a federally-declared disaster. Resources like water, food and mass care supplies are readied at the FSA for deployment to areas in need (FEMA, 2022). They are dispatched by the truckload to intermediate or final destinations in the disaster-affected area, such as shelters and other sites (FEMA, 2022; FEMA, 2019). For each load, a path must be planned using the existing road network from the FSA to the destination. These delivery efforts are typically challenging. Roads are often closed due to high water, debris or damaged infrastructure (FEMA, 2020), and these disruptions are only discovered as the response progresses. The disaster response routing literature refers to this problem as online path planning. (Here, *path planning* is distinguished from *route planning*, in that the former entails planning a path between a prescribed origin and destination pair, whereas the latter entails planning a route that begins and ends at a depot and visits multiple destinations in between).

New tools are available to support many aspects of disaster response logistics, including a variety of optimization models (e.g. Caunhye *et al.*, 2012; Gutjahr and Nolz, 2016), information technology (e.g. Özdamar and Ertem, 2015) and analytics and artificial intelligence (Spencer, 2021; De Boeck *et al.*, 2023). However, major gaps remain between the achievements of research and the problems of practice (Besiou and Van Wassenhove, 2020), and few of these tools are actually in use (e.g. IFRC, 2013). Better understanding of humanitarian practice and its challenges could help bring research into better alignment with the needs of practitioners, thereby supporting improvements in humanitarian logistics (Besiou and Van Wassenhove, 2020; Kovacs and Moshtari, 2019; de la Torre *et al.*, 2012; Kunz *et al.*, 2017).

This paper aims to investigate and compare how practice handles and how research models this disaster transportation problem – transportation between the FSA and final destinations in the presence of dynamically evolving and uncertain road conditions. Specifically, interviews with practitioners and qualitative analysis techniques are used to identify common practices for each of the following elements of this problem:

- who is doing the planning;
- how those doing the planning think about the network;
- what objective(s) guide their planning decisions; and
- how they form and update their knowledge regarding the uncertain and online problem inputs, namely, the status of roads.

In the case of the latter, this paper discusses several dimensions of the road status knowledge dynamics. We begin with information sources that planners consult to learn about road status. Next, we discuss efforts to discover new

information about road status, whether intentional or unintentional. Finally, we describe to what extent road status knowledge is shared between network users. For example, does the common knowledge base span organizations or is it relegated within an organization? Is that common knowledge base updated regularly and with intention, or instead in an *ad hoc* fashion?

As we explore each of these problem elements, we provide evidence from practice as observations of how real world actors currently operate in this decision space. We also connect those observations to the online disaster response routing literature, to illuminate how current models are similar to and different from practice. Finally, we discuss what implications for practice and modeling these similarities and differences entail.

This paper is structured as follows. Section 2 discusses relevant literature and highlights the research gap addressed in this study. Section 3 elaborates the research design. Section 4 presents and discusses the findings, comparing observations from practice to the literature on online path planning under uncertainty. Section 5 discusses the implications of our findings for improving both theory and practice in this area. Finally, Section 6 concludes the paper by summarizing key gaps between path planning models and real-world practice.

2. Literature review

Two distinct streams of literature are relevant to this paper's aims: the modeling literature that most closely represents the planning problem at the FSAs, and previous attempts to compare such theoretical models to practice. Therefore, models for online path planning in disaster response under road status uncertainty, the specific area of practice this paper investigates, are reviewed in subsection 2.1. The literature addressing comparison of research and practice in humanitarian logistics is discussed in subsection 2.2. The specific gap that the present study addresses is summarized in subsection 2.3.

2.1 Online path planning in disaster response under road status uncertainty

Early papers addressing disaster relief routing assumed deterministic planning environments (e.g. Knott, 1987; Haghani and Oh, 1996). Later studies began to explicitly model the uncertainty inherent in disaster response routing. De La Torre *et al.* (2012) identify papers modeling two types of uncertainty in routes: edges that can fail (e.g. Vitoriano *et al.*, 2009; Ukkusuri and Yushimito, 2008) and edge travel times that are uncertain (e.g. Van Hentenryck *et al.*, 2010; Mete and Zabinsky, 2010; Shen *et al.*, 2009; Rawls and Turnquist, 2010; Salmeron and Apte, 2010). In those reviewed papers, uncertainty is modeled via two-stage stochastic planning models, where plans are developed in a first stage and then uncertainty is revealed and plans adapted to account for the uncertainty in the second stage. In contrast, the decision environment motivating this paper is online, with uncertainty regarding edge failures being resolved over multiple periods and plans frequently being revisited to avoid those disruptions.

The application studied in this paper is online path planning under road status uncertainty. Online models for uncertain path planning frameworks first emerged with the introduction of the Canadian Traveler Problem (CTP) (Papadimitriou and

Yannakakis, 1991) and the Bridge Problem (Blei and Kaelbling, 1999). The CTP and the Bridge Problem both model a scenario where one must traverse a network without the benefit of *a priori* knowledge of edge failures, though probabilities of edge failure are assumed to be available in the latter. In both the CTP and the Bridge Problem, edge failures are discovered in transit, as they are encountered, and the traveler must plan a new path to avoid them. Despite the scenarios motivating these models, namely, snow on Canadian roadways and bridges damaged by storms, these models were relatively slow to be explicitly adopted in the disaster response routing literature. Instead, papers advancing CTP and related online path or route planning models mention motivating applications such as robot navigation (e.g. Lita *et al.*, 2001; Guo and Barfoot, 2019), express logistics and delivery routing in urban networks (e.g. Liao and Huang, 2014; Zhang *et al.*, 2016) and minefield countermeasures (Aksakalli *et al.*, 2016). Disaster relief distribution was mentioned alongside other applications for online path planning models in Yildirim *et al.* (2019), and was emphasized in the article title in Akbari and Shiri (2021). Recent graduate theses and dissertations focused on the application of CTP models to disaster path planning problems include Alseth (2020) and Chanchad (2023).

Much of the CTP literature emphasizes theoretical analysis. For example, online algorithms for deterministic variants of k -CTP are developed and their worst case competitive ratios are proven; k -CTP refers to the special case in which at most k edges can fail (e.g. Westphal, 2008; Xu *et al.*, 2009; Bergé & Salaün, 2019; Bender and Westphal, 2015; Demaine *et al.*, 2014; Demaine *et al.*, 2021). For the Stochastic CTP, papers prove its complexity (Fried *et al.*, 2013), provide an exact algorithm with polynomial runtime for special cases (Nikolova and Karger, 2008), and also provide an exact algorithm with nonpolynomial runtime (Aksakalli *et al.*, 2016). Given the complexity of solving large instances of the Stochastic CTP, other literature develops heuristic solution approaches and analyzes their performance via computational studies (e.g. Eyerich *et al.*, 2010; Sahin and Aksakalli, 2015; Bai *et al.*, 2018; Alseth, 2020; Chanchad, 2023).

The CTP assumes a single driver traversing a graph between a single origin and destination pair. Model variants that more closely approach the planning environment present in FSAs are emerging. At an FSA, all loads share an origin (i.e. the FSA) and are being dispatched to multiple destinations via multiple drivers. Multiagent CTP models address multiple drivers traversing a graph and possibly sharing discovered information with one another as they go. In some of these models, the drivers are all trying to reach the same destination (e.g. Zhang *et al.*, 2013; Shiri and Salman, 2017; Alseth, 2020), whereas in others, each driver has their own intended destination (e.g. Lita *et al.*, 2001; Chanchad, 2023). The latter problem, with multiple agents and multiple destinations, most closely resembles the real-world decision environment at the FSA. However, none of this literature rigorously compares these models to the problems solved in practice.

2.2 Comparisons of research and practice in humanitarian logistics

It has long been recognized that there is a significant gap between research and practice in operations management (e.g.

Sodhi and Tang, 2008; Corbett and Van Wassenhove, 1993), and humanitarian logistics is no exception. A number of papers note gaps between research and practice; for example, Kovacs and Moshtari (2019) argue for the importance of understanding real practical problems as part of the research process; and others note that few optimization models are used in practice (De Vries & Van Wassenhove 2020; Gralla and Goentzel, 2018). Many papers have stressed the importance of working on practice-driven problems (e.g. Starr and Van Wassenhove, 2014; Besiou and Van Wassenhove, 2020) and suggested strategies for doing so (e.g. Kunz *et al.*, 2017; Kovacs and Moshtari, 2019).

However, only a few papers directly compare research to practice, based on empirical data on the practice of humanitarian logistics (as we do in this paper). Besiou and Van Wassenhove (2020) compared the topics of academic papers in humanitarian logistics-related special issues to the topics of a series of practitioner conferences. They found that research topics do not align very well with practitioner concerns, and they suggest several opportunities to bring them into better alignment. De la Torre *et al.* (2012) review the literature, review practitioner reports and conduct interviews with practitioners, to compare the practice of and research on relief routing. They identify specific areas of mismatch, including the lack of research on stages beyond immediate relief; a need to better incorporate uncertainty in supply, vehicle availability and demand; an opportunity to model risk-averse behavior; and an opportunity to better demonstrate the value of routing models. A few additional papers describe practice without directly comparing it to research. Recent examples include Lewin *et al.* (2018), which uses five case studies on humanitarian logistics topics, based on 40 interviews with international logistics response practitioners, to demonstrate the essential role of supply chain management in humanitarian response; and Stumpf *et al.* (2023), which uses interviews and financial data from practice organizations to examine the importance of and costs invested in supply chain management in the humanitarian field. Both papers also identify important opportunities for future research to contribute to improvements in humanitarian practice. However, given the volume of research on humanitarian logistics – meriting 43 review papers as of 2018 (Kovacs and Moshtari, 2019) – there have been relatively few attempts to characterize practice in comparison to research.

2.3 Summary of contribution to literature

The present paper responds to the call for better matches between practice and research by examining a particular problem – online path planning in disaster response under road status uncertainty – and comparing the assumptions made in CTP models and variants to the way such problems are dealt with in practice. The limited literature comparing models with practice has not addressed this problem nor made such a detailed and systematic comparison of practice with research in a single focused setting. We draw on a series of interviews that enable us to characterize, in detail, how responders in the US disaster response system accomplish disaster relief path planning. In drawing heavily from the empirical context, we use a similar approach to that of de la Torre *et al.* (2012). However, our focus is on a narrower problem, online path planning under

road status uncertainty, and we provide a deeper understanding of practice in this area.

3. Research design

To understand current practices in disaster response path planning from FSAs in the USA, which requires an exploratory approach, we adopted an inductive qualitative research approach based on grounded theory (Corbin and Strauss, 2008). An overview of the research approach is given in Figure 1. Each of the steps in the figure is described in detail in the remainder of this section.

The first step (see Figure 1) was to define the research questions, which focused on how disaster responders planned transportation between the FSA and final destinations in the presence of uncertain road conditions. A formal interview guide was structured based on insights from an initial literature review and preliminary conversations with emergency responders. The interview guide is included in the Appendix. Due to the inductive nature of this study, the guide consisted of a set of open-ended questions focused on key aspects of disaster response transportation activities, grounds that prompt changes in transportation planning and factors that influence achievement of emergency response goals.

Next (second step in Figure 1), interviews were conducted with personnel from entities at different levels who actively participated in disaster logistics operations, especially during the 2017 and 2018 storm seasons in the USA. The primary sampling strategy was purposive: the goal was to identify participants from multiple organizations of each of several types (federal, state, private and nonprofit). The investigators' professional networks were used to identify initial participants for the interviews. Those participants suggested additional possible interviewees, following a snowball technique. Identification of participants concluded when at least two interviews for each organization type were conducted and the knowledge gained from additional interviews was small. Over the six-month period from January to June 2019, 14 in-depth interviews were performed with participants from 14 different organizations, comprised of five federal-level entities (denoted F1, F2, F3, F4 and F5), three state-level organizations (denoted S1, S2 and S3), four private companies (denoted P1, P2, P3 and P4) and two nonprofit organizations (denoted N1 and N2). (Two of these interviews, F1 and F5, were on related but different topics and did not provide information relevant to path planning within a FSA, so they are not referenced in this paper.) There were a total of 22 participants, as many of the interviews included more than one person from the participating organization. With different backgrounds and roles, these individuals provide a diverse set of perspectives that

span the relevant organizations taking part in hurricane response in 2017 and 2018.

Most of the interviews were conducted by two interviewers. One primary interviewer asked the questions while the second interviewer took notes and ensured the flow of conversation remained on track and conformed to the research objective. The duration of the interviews ranged from one to one and a half hours. Although the interviews were structured by the base questionnaire, any relevant lead was pursued with follow-up questions to reveal additional information. In all cases, the findings are presented in a manner that does not disclose information that could expose or be linked to the identity or affiliation of the respondents. Interviewees are identified only by their scope of engagement and responsibility such as local, private, state or federal. All the interviews were recorded with the consent of the interviewees for documentation. This study was conducted with Institutional Review Board approval.

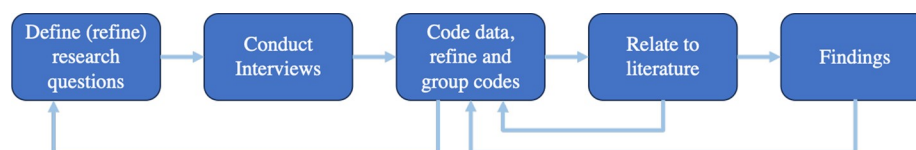
The interview audio recordings were transcribed into text using otter.ai software and checked manually to ensure proper conversion. The transcription of all interviews was completed within one month of the last interview and their lengths ranged from 4,600–14,000 words each. The interview notes and other relevant data were also documented. These data were then imported into atlas.ti software for qualitative analysis.

The third step (see Figure 1) was data analysis. The first part of the data analysis was open coding. Labels or “codes” were attached to the text reflecting a conceptual description of the corresponding content. For this paper, we coded data related to how path planning activities were accomplished. This process was completed by three researchers independently; next, the open codes were discussed among the researchers and a consensus reached about an initial set of concepts (codes) and their definitions. Once the initial codes were defined by the research team, one author proceeded to re-code the data accordingly.

The next step in data analysis was to investigate the relationships between codes and identify higher-level categories that relate them to one another (axial coding). Similarly coded segments of data were examined together, and differently coded segments were compared, to refine the code definitions and identify relationships between codes. Codes were split or combined as needed during this refinement process, and new codes were defined as new concepts emerged.

At this stage (fourth step in Figure 1), concepts from the literature were also incorporated, either by defining new codes or by identifying existing codes that referred to the same concept as the literature. Through constant comparison between the data, theoretical concepts and emerging codes, we refined the set of codes and their definitions. The first author continually re-coded the data with the updated code definitions

Figure 1 Overview of the research approach



Source(s): Authors' own work

throughout this process. Another author then coded portions of the data with the updated codes, and codes were compared and discussed among all the authors to ensure clarity and consistency. The final set of codes represent the key concepts that emerged from the data and that existed in the literature; they describe current practices in disaster response online path planning. Examining the code definitions and relationships, their frequency, the conditions under which they occur and their content, provides insights that form the basis for the findings described below (last step in Figure 1).

4. Findings and discussion

In the following sections, observations from practice and connections to literature are provided for each element of the online path planning problem under uncertainty taking place at FSAs. The findings begin with who is doing the planning (4.1), how those doing the planning think about the network (4.2) and what objective(s) guide their planning decisions (4.3). Then, how those doing the planning form and update their knowledge regarding road statuses is discussed. This includes: information sources that planners consult to learn about road status (4.4); efforts to discover new information about road status, whether intentional or unintentional (4.5); and sharing of road status knowledge between network users (4.6). Each subsection begins by describing the assumptions from CTP models in the literature. Next, the findings from the interview analysis are described. Finally, the last paragraph of each subsection compares these findings from practice to the assumptions in the modeling literature and identifies insights for modeling and practice.

In keeping with graph theory, which distinguishes between a path and a route (i.e. multistop cycle or closed loop), this section adopts the term *path planning* as defined in subsection 2.2 to describe the decision environment. This is because the interviewees are describing the selection of a path for a full truckload between a single origin and destination pair, though they refer to this as “route planning.” We keep the phrase “route planning” in interviewee quotations and use path planning in all narrative discussion.

4.1 Responsibility for path planning

Nearly all CTP models implicitly assume the driver (agent) is doing their own path planning. However, the interview data suggest that the reality is more complex. Most organizations do their own path planning centrally or leave it to a subcontracted transportation company. The path planners often coordinate with their drivers or with other organizations to deal with road closures and blockages. Table 1 shows which interviewees discussed each of these approaches to path planning.

Most of the interviewees (nine out of ten who discussed the subject) implied that a person or group within their organization was responsible for path planning. For example, one interviewee explained, “[When] helping route that load or that driver [...] our Ops [is] tracking bridges out.” [P3]. However, in most cases, they did not specify who within the organization planned paths, but instead simply referred to “we”. For example, “Trying to get 25 [vehicles] turned around in a row, we’re going to re-route them [...]” [S1]; or in a second example, “We will assess every mission as to where it’s going, which routes to take and dictate the route until the roads become more normal” [F4]. It is clear from the interviews that these organizations do their own path planning but not which part of the organization is tasked with the job. In two of the interviews, it appeared that different subteams within the organization do path planning separately. For example, one interviewee explained:

If we’re doing [one type of mission], the individual responsible for creating those [...] routes is within the [mission] team [...] And the other [mission’s] routing is dependent upon where their assignment is [...]. So there’s no national level routing methods [N2].

Three organizations noted that a private transportation company chose the paths for their own vehicles. (Transportation companies were often contracted by FEMA or other entities to transport their goods.) For example, a federal interviewee explained that:

[...] you would give one of these companies a set of loads and some delivery deadlines, and then they have the freedom or the authority to figure out the best way to do that [F3].

Four of the organizations included in the earlier category, in which path planning was done within the organization, are private entities themselves: these interviews provide additional evidence that the private companies often do their own path planning.

Some of the organizations that planned their own paths also described coordinating with local authorities. For example, one noted that “we rely heavily on TXDOT, the Texas Department of Transportation, to tell us what our options are” [S2]. Another mentioned coordinating with the state police and FEMA to determine viable paths [P2]. It is not clear from the interviews if these relationships mainly involved acquiring information from other organizations or if the path planning decisions were a collaborative process. Other organizations described coordinating with their drivers to determine paths. For example, one explained that when:

[...] you get to a road closure [that] was not on the map [or] a high water situation and you’re trying to determine how high is the water; it’s just at that point we kind of rely on the driver’s best judgment [P1].

Drivers were not described as determining their own paths, but rather coordinating with an organizational authority to re-plan when needed.

Table 1 Responsibility for path planning, according to each interviewee

Planner	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
Organization			✓	✓	✓	✓	✓	✓	✓	✓*		✓*	9
Transport company		✓							✓			✓	3
+ Coordination with drivers							✓	✓		✓		✓	3
+ Coordination with local authorities					✓			✓					2

Note(s): * indicates that the organization plans paths centrally but relies on subteams within the organization to plan paths for different teams

Source(s): Authors’ own work

Taken together, these observations suggest several relevant insights for online path planning modeling. First, many organizations do plan their own paths, suggesting that they would be able to make use of a path planning tool. However, a method for reliably communicating the path to the driver will be required, as the driver is not the entity that would use the tool. Second, there are multiple organizations working in any disaster, and each plans their own paths independently. While interviewees mentioned coordination with local authorities, there was no mention of coordination with other transporting organizations, and one interviewee even explained that such coordination did not take place [P3]. Thus, there is no overarching central authority planning paths for the disaster response as a whole. Third, on a related note, it appears that decisions on cargo prioritization are often made by a different authority than those who plan paths. Transportation companies plan paths based on required loads and delivery locations, but the decision about what to send where is not made by these companies. Thus, models that couple these decisions may be difficult to leverage in practice.

4.2 Representation of the transportation network

The literature on online path planning under road status uncertainty typically represents the transportation network as a graph, with some assumptions regarding what is known about edge status (ES) and edge travel time (ETT). Typical representations of uncertain edge status include: (ES1) edge status as available or blocked is unknown (e.g. Papadimitriou and Yannakakis, 1991); and (ES2) which extends ES1 with the assumption that a probability describing the likelihood each edge is blocked is available (e.g. Blei and Kaelbling, 1999). Regarding edge travel time (ETT), the CTP literature assumes edge travel times are known (ETT1), while other response routing models assume uncertain edge travel times described by either a stochastic distribution or a range (ETT2) (e.g. Mete and Zabinsky, 2010).

The interview data shed light on whether and how disaster response practitioners “represent” the network in their work. Table 2 summarizes which interviewees used which representations. Unsurprisingly, none of the interviewees described transportation networks in mathematical terms. Instead, most interviewees seemed to (implicitly) use road status representation ES1: they thought of roads as being open, blocked or unknown. For example, P2 explained, “The drivers tell them which routes are good and which aren’t.” Similarly, S2 mentioned talking with the Department of Transportation representative to ask, “What’s open?”. P3 reported that if a driver found a blocked road, “they call back and tell us and then our people on the ground report to FEMA and now there’s accountability that this road is blocked.” Across all the interviews, 8 of the 12 respondents showed evidence of this

representation, and the remaining four did not discuss anything related to the status of roads, suggesting that this representation is used universally.

Three respondents also spoke of roads using language similar to road status representation ES2: the *likelihood* of a road being open. For example, P2 described:

[...] if you have a road that you can’t rely on, you try not to use it [...] you go on the road you can rely on. [...] If roads have a history of having low passes and lots of water [...] they are probably going to use that tribal knowledge quickly just to close this area off because this one has a history of flooding.

Similarly, F4 explained that they:

[...] look at the risk to routes, for example a route that is a flood risk [...] we would make the assumption that’s flooded. [...] In an earthquake scenario, we study all the bridges, [...] which bridge do we think is not going to survive?

Interestingly, the three respondents using language similar to ES2 all spoke in terms of *both* representations ES1 and ES2 together: when they identified a risky route, they assumed it would be closed, as in P2’s quote above. Only one respondent used edge travel time representation ETT2, describing how they might pick a longer-distance route over a shorter one if it was likely to take less time given the traffic situation [S2].

An additional distinction evident in the interviews is whether status as blocked or available is attached to “roads” or “routes.” The terminology “road” appears roughly equivalent to an edge in a network graph, whereas a “route” is similar to a path from an origin to a destination and may comprise multiple “roads.” Table 3 summarizes which interviewees used which representations. Some organizations seem to track status on a road-by-road basis: for example, “If my driver calls in and says” “This road is a no go road” [...] they’re marking it off” [P2]. On the other hand, other organizations appear to maintain knowledge of “routes” (paths) rather than roads: they track the one or two open paths they are using, and when one is blocked, they look for a different path, rather than tracking the status of each individual road within them. For example, one interviewee explained, “[A driver] would give us a confirmed route. And then we would start shoveling all the trucks through that confirmed [route] until something changed and that route was no longer good or if we got a better route” [P1]. Another organization described using “one of those routes as our key routes to want to get into an area” [S1]. Attaching status to “routes” rather than “roads” is analogous to route and path reliability formulations in the literature (e.g. Ukkusuri and Yushimoto, 2008).

These results suggest that disaster response practitioners represent the transportation network by considering roads as blocked, open or unknown. This is consistent with literature regarding online path planning problems, like the CTP, in

Table 2 Number of discussions in each interview that used each network representation

Network representation	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
ES1: whether edges available or blocked is unknown			1	1	3	2	1	3		1		2	8
ES2: probability describing edge blockage likelihood is available			1	1				1					3
ETT2: travel time is unknown and stochastic information is available					1								1

Source(s): Authors’ own work

Table 3 Number of discussions in each interview that used each representation

Representation	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
Concept of status is attached to roads						1		1					2
Concept of status is attached to routes				1			1	2					4

Source(s): Authors' own work

which the status of edges are unknown until they have been physically observed (Papadimitriou and Yannakakis, 1991). Considering edge travel time as deterministic rather than stochastic is consistent with the CTP literature as well. Practitioners occasionally rely on additional knowledge, such as the likelihood of a road to flood, a bridge to fall or a route to become congested. However, these likelihoods are never discussed in quantitative terms (such as an 82% probability that a road will be flooded), but rather in qualitative language such as “can’t rely on” or “a flood risk” or “we think is not going to survive.” This is in contrast to developments in the CTP literature, which assume a numeric probability of failure is available for each graph edge (e.g. Aksakalli *et al.*, 2016). Furthermore, practitioners seem to operationalize knowledge of the likelihood of a road to flood by labeling it as closed when it is only “likely” to be closed or significantly delayed. This too is in contrast to the CTP literature, in which an edge is only classified as blocked if someone has observed the blockage from one of the edge’s endpoints. Finally, a “route-” or path-based representation is also different from the CTP literature and leads to different planning options than a “road-” or edge-based representation. In a road-based representation, the status of multiple roads is understood so that optimal routes can be identified. A route-based representation, however, significantly limits planning options, since planners appear to learn and “store” only the status of a handful of routes.

4.3 Goals for path planning: feasibility or optimization

The disaster response routing literature typically searches for the best, or optimal, path. This is true of models that seek to minimize travel cost or time to one destination (e.g. Eyerich *et al.*, 2010; Zhang *et al.*, 2013) and also for those that seek to minimize other objectives, such as the sum of priority-weighted distances to multiple nodes (e.g. Akbari and Shiri, 2021).

According to our interviews, this is rarely a goal of response organizations. As Table 4 shows, only three organizations appeared to be searching for the best path, and one of these was discussing a hypothetical situation. Even in these cases, the notion of “best” or “better” was not the focus. P1 explained that once they found a path that worked, they would send all their trucks along it until “that route was no longer good or if we

got a better route”, but there was no indication that they were actually searching for a better path nor what “better” means. S1 similarly mentioned a “best route,” but the core of the discussion was about finding a feasible, not better, path. S2 discussed a hypothetical idea to maintain maps that are automatically updated with path status and asset locations which could support finding “the fastest” path based on traffic conditions. In contrast, seven of the 12 organizations implied that their goal was simply to identify a workable – or feasible – path. They described many challenges with finding places where large vehicles could transit, or turn around if necessary, and identifying roads that were open. They did not mention the concept of “best” or “better” nor discuss improving paths once a feasible path had been identified (except in the rare cases mentioned earlier).

In summary, then, it appears that response organizations do not typically think about optimizing or improving their paths. Their focus, instead, is on identifying a feasible path. Few mentioned any need to speed up the delivery of aid once it had been enabled at all. This conceptualization lends itself more naturally to formulations that prioritize path reliability, similar to the most reliable path subproblem in the location routing model in Ukkusuri and Yushimoto (2008) for example, rather than the travel cost or time minimization objectives common in CTP variants. However, it can be observed that CTP solution approaches which plan a full path and follow it until a discovered disruption necessitates re-planning, as opposed to planning one-step-at-a-time the next node to visit, are already implicitly accounting for path reliability. Those approaches typically use an approximation of *expected cost to destination* to select a path, where that approximation is based on edge failure probabilities and travel times (e.g. Shiri and Salman, 2017).

4.4 Information sources for planning

Given a required load and delivery location, the information relevant for planning paths includes the status and travel times of graph edges. The literature does not concern itself much with where the relevant data comes from, but simply assumes that it either is or is not available. For example, in the CTP literature, travel times are known for all edges and the statuses of graph edges are initially not available, but become available once edge endpoints are visited and the edges are, thus, observed (e.g. Papadimitriou and Yannakakis, 1991). In Stochastic CTP models, edge failure probabilities are available for all uncertain edges (e.g. Blei and Kaelbling, 1999). In this section, we explore where information for path planning comes from, which will shed light on what data could be available for use by path planning models in disasters and when it could be available.

Interviewees described a wide variety of information sources used in path planning and re-planning. Table 5 shows the types of information sources mentioned by each respondent. The

Table 4 Interviewee reports of optimizing versus simply identifying a feasible path

Goal	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
Optimize paths				✓	*		✓						2
Find a feasible path		✓	✓	✓	✓	✓	✓	✓	✓				7

Note(s): * indicates a discussion of a hypothetical practice

Source(s): Authors' own work

Table 5 Information sources mentioned by interviewees

Source of info↓	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
Driver network							✓	✓				✓	3
Maps				✓	✓	✓	✓				✓		5
Scouts				✓			✓						2
Weather forecast						✓		✓					2
Public knowledge				✓				✓					2
DOT	✓	✓	✓	✓			✓						5
Local entities	✓					✓				✓			3
State entities	✓	✓				✓		✓	✓				5
Private parties			✓					✓	✓				3
Broadcast			✓					✓					2

Source(s): Authors' own work

driver network is an important source of information, although it was only discussed by three organizations. Drivers who are on the road, regardless of their involvement in emergency response, share information on real-time road status. Drivers communicate with one another via their radios, sharing information about both closed and open roads. For example, one interviewee explained that when their organization tells a driver that a route looks closed, the driver might say, "No, I just talked to this guy [and] he told me that that route is open" [P1]. Another explained that the drivers were "hearing on their CB radios [...] that these routes are no longer any good" [P2]. Information from the driver network, and information drivers learn simply by encountering closures, is typically relayed to their organization, so that the organization can re-plan for other drivers accordingly. Scouts, discussed by two organizations, serve a similar function of relaying real-time road status, but they are sent out purposely to assess the road condition before deploying trucks on that path. They may take the form of a drone, a person or a team.

Other responding organizations are also important sources of information. Local entities include local officials, law enforcement, sheriffs and fire and rescue teams. Similarly, state entities include state officials, troopers, police, the highway patrol, state-level highway officials and environmental officials. The Department of Transportation is a particularly important source of information, through its website, maps and deployed representatives and liaison officers, who update information about accessible roads to emergency responders. Private parties such as commercial industries and transporters using the same roads also collaborate to share information on road conditions.

Public sources are also essential sources of information, including maps, weather forecasts, broadcast radio or television news, social media posts and other sources of public knowledge.

Few of these information sources were described as providing information on the "probability" or "likelihood" that a road is open; instead, they largely provide real-time status updates on whether roads are open or closed. In the three cases in which "likelihood" information was used (see Table 2, row labeled "ES2"), it came from either historical knowledge or preparedness work, such as examining which bridges are likely to fail in an earthquake and which roads are likely to flood in a storm.

Organizations do not all rely on the same sources of information. Table 5 suggests that federal entities primarily rely on official information from different tiers of organizations. All

state entities use maps while supplementing their information with a variety of sources. Private entities in particular described a large number and variety of information sources used for path planning.

In the online path planning literature, road status information sources other than driver-encountered disruptions are relevant to specific variants of the classic CTP: CTP-sensing and multiagent CTP models. Information from other responding organizations, public sources and scouts (drone, person, etc.) can be modeled by CTP-sensing variants, where sensing costs and budgets represent the costs associated with retrieving information from those sources and the resources available for information-gathering efforts (Bnaya *et al.*, 2009). Heterogenous sensing costs that depend on information source would be a reasonable and practical extension to the literature. Sensing delays may also need to be introduced in models to capture time required to collect and disseminate remotely sensed information. Information from driver networks can be modeled by multiagent CTP variants, where drivers send and receive information to and from one another (e.g. Zhang *et al.*, 2013).

4.5 Discovering the network status

The disaster response routing literature makes assumptions about how the network status is discovered. In models for online path planning along disrupted networks, such as the CTP, network status is initially unknown and is discovered as a driver traverses the network and encounters roads (e.g. Papadimitriou and Yannakakis, 1991). In CTP variants that allow for communication between drivers, an individual driver may discover network status via their own observations and also via communication of information discovered by other drivers (e.g. Zhang *et al.*, 2013). Other models, such as CTP-sensing, allow for discovering the status of some network edges without visiting them, sometimes at a sensing cost (Bnaya *et al.*, 2009; Zhang *et al.*, 2016). Finally, some models allow for network status discovery through Bayesian inference, where a driver can infer the status of an unvisited road based on observations of correlated roads (Lim *et al.*, 2017).

All papers reviewed in the prior paragraph provide models in which drivers are not intentionally trying to learn about the network status; rather, they are simply driving along planned paths and they learn the status of included roads as they are encountered. In contrast, one might characterize the model in Zhang *et al.* (2015) as intentional exploration of a network to find a fast and unblocked path to destination. Though the authors do not frame their model in exactly that light, the idea is to determine a set of multiple short and dissimilar vehicle paths between a shared origin and destination that enable at least one vehicle to reach the destination quickly. A similar formulation is the k -Canadian Travelers Problem with communication, where k drivers sharing an origin and destination share discovered information with one another as the network is traversed, and the shared information can be used to replan when blockages are encountered (Zhang *et al.*, 2013).

In practice, a wide variety of the above methods enable discovery of the network status. Table 6 shows the different types of network discovery approaches identified in the data. By far the most common is through communication with another entity, usually either a driver or an organization that aggregates

Table 6 Interviewee reports of learning about the network status

Discovering network status. . .	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
By exploring, unintentionally					✓	✓	✓	✓		✓		✓	6
By exploring, intentionally						✓							1
Through communication		✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	10
By sensing at a distance (e.g. . . .)		✓		✓	*		✓	*	*				3
By inferring based on other information (e.g. history, road type, etc.)			✓	✓				✓					3

Note(s): * indicates a discussion of a hypothetical practice

Source(s): Authors' own work

information like DOT or FEMA (see subsection 4.6). Ten of the 12 organizations mentioned this discovery method.

The next-most-common approach (described by six of the 12 organizations) was to learn from drivers who encounter closed roads: we term this “unintentional exploration.” For example, as an interviewee from P1 described, “you’re coming down a country road, you’re turning left and [. . .] then you get to a road closure that was not on the map.” We distinguish this unintentional exploration from “intentional exploration,” in which an organization sends someone (such as a police officer or a member of an advance team) to drive the network in a deliberate attempt to discover the status of roads. Only one of the 12 organizations described such an approach.

Inferring road status based on other information was described by only three organizations. Interviewees described making assumptions about “which bridge do we think [is] not going to survive” [F4] or to “close this area off because this [road] has a history of flooding” [P2]. In all cases, the inferences were based on knowledge of a road’s infrastructure or history, such as whether it contains bridges or has flooded in the past.

Learning road status by sensing it at a distance was described by only three organizations, but was also mentioned by three additional organizations as a strategy they would like to use. Many different methods of sensing were mentioned. Those already using this strategy described a “social media monitor” looking for things like “hashtag I-35 is closed” [F3] and visual imagery from aerial vehicles [P1 and S1]. Those discussing hypothetical ideas talked about live tracking for assets and infrastructure status, informed by artificial intelligence and mapping technologies.

In sum, most organizations discover the status of the road network in *ad hoc* ways: primarily either as a driver explores the network or through information communicated by other entities, discussed further in subsection 4.6. The former is consistent with CTP models in the literature that assume a single driver on the network; the latter is consistent with CTP models that assume multiple drivers on the network sharing information with one another (e.g. Zhang *et al.*, 2013; Shiri and Salman, 2017). Very few organizations currently leverage the sensor capabilities assumed in the CTP-sensing model or inference to estimate the network status as assumed in the Bayesian CTP (Bnaya *et al.*, 2009; Zhang *et al.*, 2016; Lim *et al.*, 2017).

4.6 Knowledge sharing

In online path planning models in the disaster relief distribution transportation literature, edge information is discovered as a

graph is traversed. For models that include more than one agent (i.e. driver) moving along the graph, how information is shared among those agents is addressed. For example, in Bergé *et al.* (2019), agents communicate with one another during planning only at the source node. Zhang *et al.* (2013) introduce a multiagent CTP variant with two communication levels – *full communication*, in which RS-type agents both receive and send information as they traverse a graph, and *limited communication*, in which R-type agents can only Receive information. Shiri and Salman (2017) introduce three communication protocols for RS-type agents, which differentiate between only sharing discovered edge blockages (CP1), additionally sharing travel information, which implies traversed edges are confirmed to be available (CP2) and making decisions for (i.e. planning paths for) R-type agents (CP3). Shiri and Salman (2017) also differentiate between three intelligence levels, in which an agent can only make a decision about their own next travel move (IL1), can plan their own full path (IL2) and can plan and transmit another agent’s full path (IL3). In all of the papers reviewed in this paragraph, any communication that is allowed is assumed to happen instantaneously, as soon as an agent reaches an edge endpoint to discover its status. It is implied that all information shared “out” or “up” by agents capable of transmitting information is maintained in a central repository that describes what is globally known about the status of graph edges at any point in time. In this framework, disruptions discovered locally by R-type agents (who are unable to send information) will not be globally known. Lita *et al.* (2001) introduces a multiagent, multidestination CTP variant where information agents, who are trying to reach a destination, send collected information to a central planner, referred to as a dispatcher agent. The dispatcher agent considers the global repository of discovered information to update plans and transmit them to information agents.

As online path planning models with multiple agents is a developing area in the disaster response literature, this section is framed in a way to provide insights to model development from the practitioner perspective.

One important question is whether there is a formal knowledge repository and whether it is maintained by an organization or shared across multiple organizations. The first two rows of Table 7 summarize the results. Only one of the interviews describes a formal “common operating picture:” a map or other information repository that captures information from many sources about road and infrastructure status. This is managed at the federal level by the situational awareness branch [F4], and includes information from commercial providers, states, law enforcement and other sources. However,

Table 7 Interviewee reports about road status knowledge repositories

Knowledge repository/update type	F2	F3	F4	S1	S2	S3	P1	P2	P3	P4	N1	N2	Tot
Inter-org knowledge			✓	✓		✓		✓				✓	5
Within org knowledge			✓				✓	✓					3
Ad hoc updates		✓				✓	✓	✓			✓		5
Regular updates			✓			✓							2
Person available		✓	✓	✓									3
Digital knowledge updates													0

Source(s): Authors' own work

some evidence suggests that this federally maintained knowledge is often too far from the action to be up to date.

Most of the other interviewees refer informally to knowledge of road or route status but do not describe any formal repository in which it is maintained. The two private organizations seem to store and update this information internally; for example, one interviewee states, “You try to keep an accurate as possible go no go route,” [P2] implying it has an internal knowledge repository, but does not describe how this is accomplished nor what form it takes. More often, organizations seem to update a repository maintained by another organization. For example, the same private organization also describes learning from the Department of Transportation, drivers and state troopers about the status of roads, and then “call and tell [FEMA] headquarters” [P2]. Other interviewees describe a similar process of updating another organization’s knowledge, but the repository is maintained by a state entity [S1] or law enforcement [S3] or a transportation company [N2].

Thus, it appears that significant inter-organizational coordination takes place regarding a “common operating picture,” but that the process is not systematic nor consistent across organizations and disasters. There is no single knowledge repository that is updated or used by these organizations, despite the interviews’ focus on a single geographic area and disaster. A federal entity maintains a central knowledge repository, but it is far from the action and may not always be up to date, and it does not appear that the private organizations, who do much of the path planning, rely heavily on it. Furthermore, there is no evidence of any kind of easily shareable, digital representation of road status knowledge.

A related question is whether there is a systematic process for sharing and updating information. The last four rows of Table 7 summarize these results. Many organizations appear to generate or request updates in an *ad hoc* manner, driven by newly acquired information or an arising need for information. For example, one interviewee explained, “if my driver calls in and says this road is a no go road, and I call it in higher” [P2]. Another noted that they might be “looking at a map and saying” “it doesn’t look like that’s open” but [someone might say] “no, I just talked to this guy. And you know, he told me that that route is open” [P1]. In many cases, information does not flow as quickly as it could, because the organizations with information to share may have “a hundred other battles they’re trying to work with” [P2]. There was, on the other hand, very little evidence of regularly scheduled calls or meetings to update information. Only two organizations described something close to such a process: the federal entity maintaining the “common operating picture” described a goal of learning from all

knowledgeable entities but did not describe a systematic process for doing so [F4], and a state entity mentioned a weather forecast authority “feeding us” information, but again did not describe the process itself [S3]. A slightly more common, but still uncommon, arrangement was a person located within or near the organization who had access to updated knowledge. For example, one organization mentioned “a representative from the Department of Transportation sitting here monitoring all the routes and telling us what’s really open and closed” [F4]. No organizations mentioned any kind of digital knowledge updates, such as logging into a mapping tool like Waze or updating a GIS map.

Thus, there is clear evidence of *ad hoc* knowledge sharing via calls, particularly driven by the acquisition of new information or a new need for information. However, new information is not always shared immediately when acquired, but may only be shared when a request is made. The private and nonprofit organizations, in particular, appear to rely on this type of process. Outside of this *ad hoc* process, there is almost no evidence of a systematic process such as regularly scheduled updates or a digital tool for information sharing. The federal and state organizations appear to aspire to such a process but did not provide details. Since the private organizations make many of the routing decisions (as discussed above), it appears that the *ad hoc* process drives the information available to path planners.

In summary, information sharing in practice does not seem to be as thorough as multiagent online path planning models assume. That is, there is evidence of some edge blockages being shared *ad hoc*, but there are not systems in place to ensure every discovered blockage is communicated to all graph users nor stored in a central repository for all graph users to access. This mimics the partial communication scenario in Zhang *et al.* (2013) where not all agents traversing the network are able to transmit discovered information. Furthermore, the sharing of new information about the graph does not happen instantaneously as multiple models assume (e.g. Lita *et al.*, 2001; Zhang *et al.*, 2013; Shiri and Salman, 2017). Instead, a series of delays impact information sharing, as the driver must relay the information, it must be updated to a central repository, and then another graph user eventually accesses the information.

5. Implications for research and practice

This section considers the implications of key observations comparing path planning practices at FSAs to modeling for online path planning on uncertain road networks. Section 4

developed these observations. To summarize, first, while most CTP models assume the driver does their own path planning, our evidence shows that a central authority within each organization typically plans paths (4.1). Second, while research and practice both represent the transportation network by labeling roads as blocked, open or unknown, they represent uncertainty in very different terms (4.2). Third, researchers typically seek the fastest feasible paths, while response organizations typically seek feasible and/or reliable paths (4.3). Fourth, the information sources used in practice vary widely, and researchers have modeled many of them (4.4). Fifth, models have captured multiple ways that response organizations discover the status of the transportation network, as well as some methods that are not used in practice (4.5). Sixth, information sharing in practice is *ad hoc*, whereas models tend to assume information is shared thoroughly and instantaneously (4.6). Based on these gaps, the following discussion elucidates both how models can better meet practitioners where they are and opportunities for practice to leverage modeling advances.

5.1 Network status

For researchers, our findings suggest a need for changes in assumptions on path planning centralization and communication. State of the practice does not support that algorithms for planning would be adopted at the driver level, via onboard computing systems available in the vehicle. Rather, it is more likely that any algorithm for planning would be used by a central planner in the driver's organization. Therefore, researchers should assume centralized planning by each organization, not by spatially distributed drivers nor across the entire multiorganization response. In addition, our findings suggest that communication between the driver and planner is reliable but basic (i.e. voice-based). While the disaster response literature establishes the unreliability of some communication channels during a response, such as cellular and wi-fi (e.g. Ali *et al.*, 2015), our data show that the use of more primitive technologies such as CB radio fill the communication gap. Therefore, algorithms should not assume rich technical communication between drivers and central planners.

For researchers, our findings highlight the partial or delayed sharing of network information in practice. CTP variants either use a driver-as-planner framework that does not require communication at all, as in the single agent models, or make assumptions regarding communication that are unrealistic when compared with current practice, as is the case with multiple agent model variants. Regardless of where the degree of communication in a multiple agent CTP model falls on the limited-to-full spectrum, the communication that does occur is assumed to be instantaneous. That is, a driver transmits information regarding a failed edge as soon as they arrive at an endpoint of the edge, and that information becomes available to other drivers as soon as they have a need to replan. Some models additionally assume information for edges confirmed to be available is also communicated with this level of immediacy. In reality, there are delays associated with a driver calling in a failure to a central planner. And, not all drivers on the network have access to the knowledge newly acquired by the central planner, especially if they are part of a different organization. Whether drivers from a second organization can benefit from the new knowledge depends on:

- hearing it directly in channels among driver networks (e.g. CB radios) and calling it in to their own planner; or
- the knowledge is conveyed by the first organization to a higher-level authority, e.g. the state DOT, and that authority disseminates the information back out in a way the planner at the second organization can access it.

To more closely resemble practice, researchers should model this partial sharing of information more explicitly in CTP variants, and should also introduce delays for diffusion of new road status knowledge. Models such as CTP-Sensing should also be further developed to account for new road status knowledge being made available over time from sources other than driver updates alone, such as from drone surveillance and news reports.

For practitioners, our findings demonstrate the need for a common operating picture of network information. To facilitate faster and more complete diffusion of road status updates to *all* network users, systems and processes governing the formation of a common operating picture for road status are required. Furthermore, any adoption of planning algorithms from the literature would require reliable communication of such information between the driver and the planner – for the driver to send road status updates to the planner, and for the planner to send path updates to the driver. The disaster management literature has long recognized the need for real-time communication and shared collections of data available to all organizations operating in response (Mendonca *et al.*, 2000; Lettieri *et al.*, 2009). Relational databases could be used to link the road status information available to various organizations (Lettieri *et al.*, 2009). A visual representation of the collected data, for example, in a geographic information system (GIS), would be especially useful (Mendonca *et al.*, 2000). Current GIS systems (e.g. Google Maps) do provide a “sunny-day” snapshot of road infrastructure in an area. What is missing from these systems is layers describing edge failure probabilities and up-to-date road status information, shared across all system users. Furthermore, if the road attribute layers are to be used in a central path planning algorithm, the GIS and any planning algorithms should be interoperable.

5.2 Stochastic information pertaining to network status

For researchers, our findings demonstrate a need for alternative models of uncertainty in network edge status. Much of the CTP literature focuses on the development of stochastic rather than deterministic models and demonstrates that the consideration of edge failure probabilities improves travel cost, i.e. reduces the time required for a driver to reach their destination. For example, the best stochastic policy in Eyerich *et al.* (2010) outperforms the deterministic policy included in the paper by 17% to 20% for test instances in their computational study. In contrast to the online path planning literature, our evidence shows that current planning practice is deterministic in most cases; the exceptions are when an organization considers failure likelihoods for only some edges, and further, completely excludes those “possible-to-fail” roads from planned paths. No models in the literature capture this particular treatment of stochastic information, where edge probabilities are only available for some edges, and edges which can fail are assumed to be failed until proven otherwise.

Development of such models would allow for a comparison of current practice to state-of-the-art stochastic models. Probabilistic approaches show potential, though more evaluation on real-world case studies is needed.

For practice, our findings suggest the importance of integrating probabilistic data for network edges into GIS systems. Stochastic models require failure probabilities for all edges that are subject to failure. Practice suggests that failure probabilities are only available for some edges, and may not be precisely quantified. That is, rather than describing an edge as having an 82% chance of failure, edges might be described as having a history of flooding. Realizing the potential benefits of existing stochastic models will require quantifying uncertain edge inputs, such as measuring a road's propensity for flooding or a bridge's likelihood of falling on a continuous [0,1] scale. Data are available to support numerical modeling efforts, such as the general condition data in the National Bridge Inventory and inundation predictions from the Super-Fast INundation of CoastS compound flooding model (U.S. Department of Transportation (USDOT), 2008; Leijnse *et al.*, 2021). As the work of integrating these probabilistic data into shared GIS systems is time-intensive, the integration should be completed pre-disaster if the data is to be leveraged in a response.

5.3 Goals for path planning

For researchers, our findings suggest the need to model responders' reliability goals for planned paths. The results presented in this paper indicate that planners currently prioritize path reliability over travel cost or time. Existing stochastic models for CTP indirectly address reliability with their minimization of expected travel cost. This objective function will bias against selection of paths that include edges with high probabilities of failure, as any encountered failures require some backtracking (and, thus, increased travel) on behalf of the driver. However, the focus in practice on reliability suggests online path planning models should treat reliability more explicitly in future research.

5.4 Limitations

This study's primary limitation stems from our focus on a single set of organizations encountering the CTP-like problem in the context of FSAs for hurricanes in the USA. It is possible that responders in other countries, or in the international humanitarian community, approach this problem differently or have different goals, tools or capacities. Future work could use similar methods to explore how practitioners in different settings approach the same problem, or examine different logistics problems within the broader humanitarian logistics space.

6. Conclusions

The differences between the practices at FSAs and the assumptions in online path planning models explain why practice has not been able to leverage the relevant research. First, the models assume information is available which is not actually available. For example, there are no edge failure probabilities available from the start of the response. Moreover, there do not appear to be formal (e.g. digitized) repositories of edge status information at all, which would be necessary to

provide inputs to any of the path planning models. Second, models often assume a planning scenario that does not match that of practice. That is, most models assume the driver independently plans and re-plans their own paths, which was never the case in the organizations with whom we spoke. Instead, authorities within each organization plan paths for the drivers and support any required re-planning. Third, the goals of disaster responders appear different from those represented in models: they do not appear to seek improvements but rather to identify a reliable path.

To better leverage the advances of research, practice could adopt processes and technologies for sharing real-time road status information and incorporate edge failure probabilities from preparedness efforts in those systems. Currently, while road status information is shared, it is done in an *ad hoc* manner, and does not appear to be maintained in a database or system which could provide the data needed to leverage modeling advances.

To enable adoption of innovations in practice, researchers could model information diffusion delays and evaluate alternative treatments of reliability and stochastic edge status information. Such adaptations could bring research into better alignment with the constraints and goals of disaster response practitioners. Research must acknowledge and adapt to the limitations of data availability, communication and the human and technological capacity to run sophisticated models during a disaster response, and ensure that models align with the goals of response planners in focusing on reliability before speed.

While this investigation focused on a narrowly defined situation – path planning from a staging area to a final destination during a disaster response – the conclusions may be more broadly applicable. Disaster response logistics is often plagued by uncertainty about infrastructure, supply and demand, such that planning must be updated continuously as information evolves. Practitioners often lack systems for collecting, maintaining and sharing data in real time, which inhibits not only the response but also the potential to leverage decision support tools based on optimization or artificial intelligence. Thus, in a wide variety of disaster response logistics problems, researchers should be cautious to ensure that models align with practice goals and work within data and technology limitations – or demonstrate the potential improvements from making an investment in easing these limitations. Better alignment between practice and research could lead to significant improvements in response speed and effectiveness, easing the suffering of those affected by disasters.

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Further reading

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Appendix. Interview guide

- When responding to disasters, what is the scope of your organization's responsibilities and the authority you have to deploy resources?
- What are the three most recent events that you have needed to respond to, and their relative magnitudes? Please

elaborate on level of resources (personnel, equipment, supplies, capital, etc.) deployed in response to this event.

- What type of transportation activities were required to achieve your response goals in the events described in the prior questions? Please summarize different types of vehicles and personnel required.
- What type of transportation plans were developed in advance of this event to achieve your response goals?
- How often were transportation plans revisited, before and during the event (hourly/daily/weekly)?
- What types of changes were made to transportation plans (Routes, loads delivered, etc.)?
- In the course of responding to this event, how did plans need to be modified in light of the situation? For example, were roads blocked by traffic or event related damage?
- Which response goals were accomplished? Please describe how your transportation plan enabled this outcome.
- What tools or information increased your capacity to respond to the events? Would you change these in any way?
- Are there any tools or information which, if you had or were more effective, could have helped achieve your response goals?
- What constraints restricted your organization's response in terms of overall capacity, delays in deployment or increased response times?
- How did you overcome these constraints?
- Where are the demand points, how many of the points exist, how many shipments were each receiving, etc.?
- On what timeline are truckload shipments to the demand points happening (24 hours after? 48?) How did uncertainty (especially on road status) impact these shipments which group makes decisions in terms of what demand points needed shipments and the relative priorities among them?
- Once a decision was made to send a shipment to a demand point, what operational details made that happen (e.g. how were trucks and drivers matched to loads, who "owned" the transportation resources, etc.)
- Is there anything else you would like to add?

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